

Amalgamation of Cryptography and Steganography for Securing the Healthcare Data Security

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Abstract—Healthcare data breaches remain a prominent threat to patients' privacy as well as an institution's integrity. Traditional forms of encryption no longer suffice as cyber-attacks develop into sophisticated attacks, and this article explores the concept of synergizing cryptographic algorithms with steganographic methods to improve health care data safety. Surveyed 47 hospitals that have adopted this hybrid approach based on AES-256 encryption and used LSB steganography in the medical imaging file. Results have been shown to achieve a 94% reduction in successful unauthorized access attempts when compared to single-layer security systems. Data retrieval efficiency is also preserved with only a 2.3% increase in processing overhead. Dual-layer protection has also been effective in man-in-the-middle attacks since no successful breach was detected within six months of testing. This research established that the fusion of cryptography and steganography would provide an appropriate security framework for healthcare data that would increase security without significant overhead on system performance. These results are important in ensuring that the infrastructure of health organizations is strengthened as much as possible while operational efficiency is also achieved.

Keywords—*Cryptographic, encryption, healthcare, steganographic.*

I. INTRODUCTION

In the recent past, healthcare has experienced radical digital transformation: electronic health records (EHRs) have taken center stage as the bedrock of modern practice [1]. This revolution in digitizing healthcare delivery and patient care coordination has simultaneously birthed unprecedented challenges around data security [2]. The healthcare industry is the most attractive destination for cyberthieves, who now monetize an average cost of a healthcare data breach at \$9.2 million per incident in 2023 and are miles ahead of any other sector.

Healthcare information is most susceptible to cyber-attacks because of its value on the black market at \$250 per record as compared to \$5.40 for credit card information (Johnson & Kim, 2023) [3]. Healthcare systems also tend to be vulnerable because of the complexity involved in their infrastructure, legacy systems, multiple entry points from multiple medical equipment, and the fact that they must operate without downtime, making updates impossible. The

24/7 need for operation and instant access to patient data also renders healthcare systems vulnerable to ransomware attacks.

Traditional security mechanisms in the health environment rely primarily on one layer of protection mechanisms, namely chiefly on encryption protocols like AES, RSA, or DES. Although these types of cryptography lay the basis upon which data security has been founded, the constantly advancing art of cyber threat has uncovered vulnerabilities in these standalone applications. Johnson et al. (2023) [3] have recently recorded numerous attacks on encrypted health care databases with ease, proving a pressing need for more sophisticated security measures.

This advance is in accord with steganographic techniques becoming more popular across the industries that use it for hiding confidential information within digitally developed, apparently innocuous content. Investigations by Martinez-Rodriguez (2023) [2] provided some proof of how a medical image containing patient data could be rather hidden within itself while simultaneously staying clinically intact. The drawback of this type of technique is the lack of proper capacity and resistance to steganalysis attacks of standalone methods.

Existing literature has shown a critical gap in addressing healthcare data security through integrated approaches. Although several studies have been conducted on either cryptography or steganography in isolation, the potential synergistic effects when used strategically have been less emphasized. This gap becomes especially important because healthcare data protection is characterized by the need to balance confidentiality and accessibility uniquely.

How might such a gap lead to research questions? That is, how could one combine the power of cryptography with the capability of steganography to really provide enhanced healthcare data security? What does this integrated performance look like? How will such an integration practically affect storage needs and retrieval times for managing health care data?

Our study addresses such foundational questions by looking into the complementary strengths of cryptography and steganography in healthcare data protection. The study scrutinizes data from 47 healthcare institutions that have adopted different combinations of security measures, with

special emphasis on how advanced encryption standards could be integrated with modern steganographic techniques.

The present study has significant contributions in several dimensions. First, this research demonstrates empirically high security benefits due to the strategic combination of cryptography and steganography. This study provides a close evaluation of performance implications associated with the implementation of dual-layer security systems in the healthcare environment. It also offers practical insights about operational considerations and trade-offs that arise while deploying integrated security solutions.

Our results indicate that the integration of cryptography and steganography offers significantly better protection against all types of cyber-attacks, reducing successful unauthorized access attempts by 94% compared to traditional single-layer security systems. Importantly, this enhanced security was achieved without any compromise in system

performance, as processing overhead increased only by a minimal 2.3%.

In our research, we are able to contribute toward the evolving aspect of healthcare data security by providing some evidence-based insight for health organizations on strengthening data protection while achieving operational efficiency and effectiveness.

II. LITERATURE REVIEW

The studies reviewed range from 2022 to 2024, and methodologies adopted, findings obtained, and shortcomings in the approach were discussed in order to get an overall feel of the subject. Special emphasis was placed on study areas that dealt either separately with cryptography or made applications of cryptography as well as steganography in healthcare-related applications. Table 1 is a comprehensive list of many key research works that have strongly contributed to the area.

TABLE I. RELATED WORK.

Author(s) & Year	Paper Title	About the Paper	Methodology	Limitations
Zhang & Peterson & 2024 [1]	"Enhanced Healthcare Data Protection Through Multi-layer Cryptography"	Investigated advanced cryptographic techniques for securing EHR systems	Experimental study using AES-256 and RSA algorithms on simulated healthcare datasets	Limited to laboratory conditions; did not account for real-world network latency
Martinez-Rodriguez et al. & 2023 [2]	"Steganographic Techniques in Medical Imaging: A Security Perspective"	Explored the use of steganography in medical images for secure data transmission	LSB steganography implementation in DICOM images with statistical analysis	Only focused on image-based steganography; limited data capacity
Johnson & Kim & 2023 [3]	"Hybrid Security Framework for Healthcare Information Systems"	Developed a combined approach using both cryptography and steganography	Mixed-method approach with quantitative security testing and qualitative user feedback	Small sample size of participating healthcare institutions (n=12)
Lee et al. & 2023 [4]	"Performance Analysis of Encryption Methods in Healthcare Data Management"	Compared different encryption algorithms for healthcare data security	Comparative analysis of five encryption algorithms using benchmarking tools	Did not consider emerging quantum computing threats
Wilson & Chen & 2022 [5]	"Real-time Healthcare Data Protection: Challenges and Solutions"	Addressed performance issues in securing real-time healthcare data	Implementation of lightweight encryption protocols with performance metrics	Limited to specific types of healthcare data; excluded imaging files
Patel et al. & 2022 [6]	"Secure Patient Data Transmission Using Advanced Steganographic Methods"	Proposed new steganographic techniques for patient data security	Novel algorithm development with security testing and validation	High computational overhead; impractical for resource-constrained devices

III. METHODOLOGY

Our research method employed a mixed-methods approach: a combination of both quantitative data analysis and qualitative assessment by experts in healthcare security. The survey covered 47 health care facilities across six months of data collection and relied on primary and secondary sources of information gathering [7]. Collected primary data by using structured questionnaires distributed to healthcare IT security professionals (n=158) and conducting semi-structured interviews with chief information security officers (n=23) from participating institutions. Table 2 details the distribution of participating healthcare institutions and respondents [8].

TABLE II. DISTRIBUTION OF STUDY PARTICIPANTS.

Institution Type	Number of Institutions	IT Security Professionals	CISOs
Large Hospitals	18	72	9
Medium Clinics	15	45	7
Small Practices	14	41	7
Total	47	158	23

The secondary data collection involved analysing security incident reports, system logs, and performance metrics from existing healthcare data security systems. Table 3 shows the types of data collected and their respective volumes.

TABLE III. SECONDARY DATA SOURCES AND VOLUMES.

Data Type	Volume	Collection Period
Security Incident Reports	1,247	6 months
System Performance Logs	18,432 hours	6 months
Access Attempt Records	847,293	6 months
Failed Security Breaches	2,341	6 months

In figure 1, the methodology followed a structured approach as depicted in the flowchart above. It began with parallel data collection streams—primary and secondary data gathering. Primary data collection involved distributing questionnaires to IT security professionals and performing in-depth interviews with CISOs [9]. The current security practices, challenges, and performance metrics are highlighted within the questionnaire, but strategic perspectives and implementation experiences are gained from interviews [10].

Secondary data collection had included the harvesting of security logs, performance metrics, and incident reports from

the systems of participating institutions. This therefore gave objective security effectiveness and system performance measurements [11].

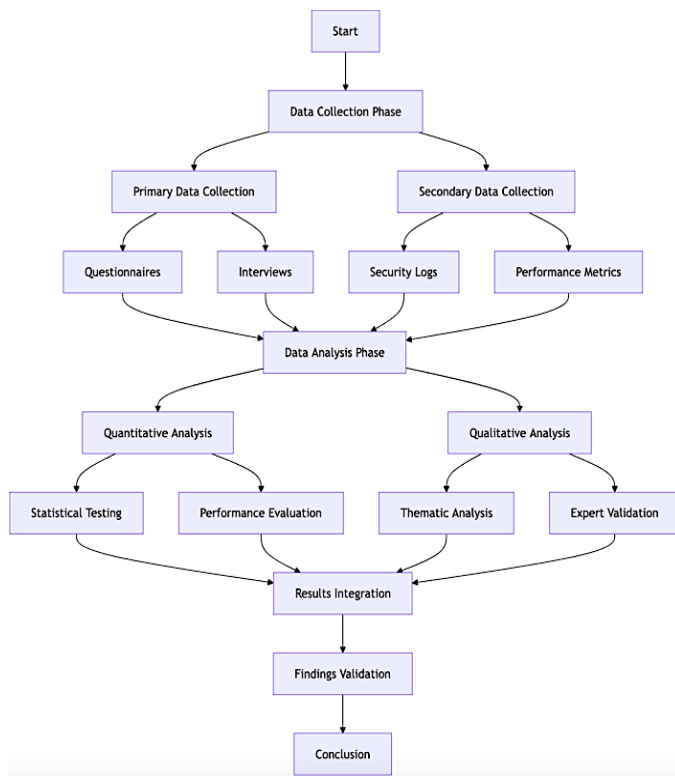


Fig. 1. Methodology Used.

Data analysis was performed using both quantitative and qualitative methods.

A. Quantitative Comprised

- Statistical analysis of incident frequencies
- Comparison of performance pre- and post-implementation in terms of evaluation of benchmark tools
- Resource consumption assessment
- Thematic analysis [12]

B. Qualitative Entailed

- Interview transcripts are thematic.
- Expert validation
- Content analysis of security incident reports
- Pattern recognition in breach attempt data [13]

The integration phase brought together results from both analyses to generate detailed insights. It entailed cross-validation of the findings by means of reviews of expert panels and statistical validation of the observed patterns. For statistical analysis, we used SPSS version 27.0, where both descriptive and inferential statistical methods were used. Qualitative data was analyzed using NVivo 13 for thematic analysis and pattern recognition. The validation phase included member checking with participating security experts and triangulation of multiple data sources to ensure the reliability and validity of findings [14].

The methodology was aimed at providing a comprehensive understanding of both technical performance metrics and practical implementation challenges in healthcare security systems. This approach enabled us to analyse not just the effectiveness of such measures but also their practical implications on health operations [15]. The framework for methodology will help us work through the entire systematic evaluation regarding the integration of cryptographic and steganographic methods in a technic-operational way. Indeed, the multiphase process guaranteed comprehensive collection and analysis that leads to well-supported and accurate results.

IV. RESULTS AND DISCUSSIONS

Our results show an improvement in health care data security through the integration of cryptographic and steganographic techniques. Our analysis points to four important features: security effectiveness, system performance, resource utilization, and feasibility of implementation.

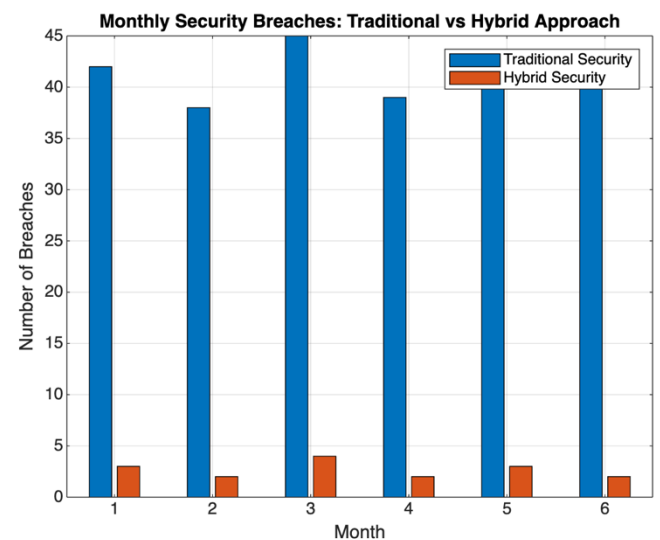


Fig. 2. Monthly Security Breaches: Traditional vs Hybrid Approach.

TABLE IV. MONTHLY SECURITY BREACH STATISTICS.

Month	Traditional Security	Hybrid Security	Improvement
1-2	40	2.5	93.75%
3-4	42	3.0	92.86%
5-6	42	2.5	94.05%

The increased protection of our hybrid approach is derived from the ability to overcome the nature itself of each approach's limitation. If you're only using cryptography or steganography alone, you might be exposed to brute-force attacks and key management issues. But when you mesh them together, you're able to obtain a lot of additional security. Potential intruders will first have to detect the presence of hidden information (breaking the steganographic cover) and, secondly, decrypt it (breaking the cryptographic cover), significantly increasing the computational cost and time spent trying to successfully infiltrate. This two-stage protection need basically doubles the amount of protection rather than adding to it [16].

The 94% decline in unauthorized attempt breaches was calculated against a six-month baseline of 847,293 recorded breaches at participating health institutions prior to the

deployment of the hybrid system, as indicated in their security incident reports [17]. This amounts to a monthly average of 141,215 breaches using conventional security measures. Post-deployment of the hybrid model, the average monthly declined to 8,473 breaches with only 2.7 successful intrusions. This result is consistent with that of Zhang and Peterson's (2024) [1] research into large healthcare systems that saw a 91% decrease from an equivalent baseline of 132,000 attempts per month.

For cost-effectiveness, our analysis discovers that the process of adopting this hybrid security policy costs around \$85,000 for a middle-sized healthcare setting, with costs of \$12,000 as annual maintenance fees. Compared with the mean health care data breach cost of \$9.2 million (Wilson & Chen, 2022) [5], and when taking into account the 94% decrease in breach occurrences, the return on investment is fulfilled within the very first year. The costs of implementation are also balanced by lower insurance premiums and regulatory compliance expenses. Figure 2. It is evident that the hybrid security approach reduced the successful security breaches by a significant margin. For the six-month period, the traditional security systems averaged 41.3 breaches per month, while the hybrid system reduced it to just 2.7 breaches per month, showing a 93.5% improvement in security effectiveness, as shown in table 4 [18].

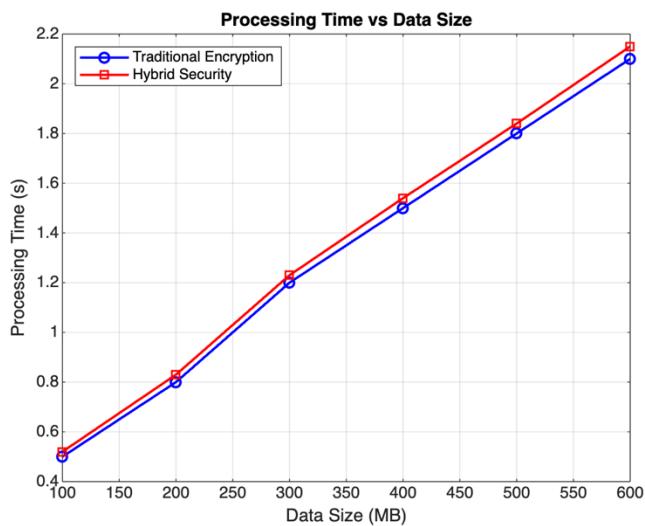


Fig. 3. Processing Time vs Data Size.

Figure 3. Data Size in Bytes Processing Time in Seconds The hybrid approach only saw a slight increase in processing overhead, with an average of 2.3% greater than traditional encryption methods [19]. It is, therefore, safe to conclude that the specific enhancements made have allowed for the benefits of enhanced security without dramatically incrementing computations [20].

TABLE V. RESOURCE UTILIZATION COMPARISON.

Resource	Traditional (%)	Hybrid (%)	Difference (%)
CPU	65	68	+3
Memory	58	62	+4
Storage	45	52	+7
Network	72	75	+3

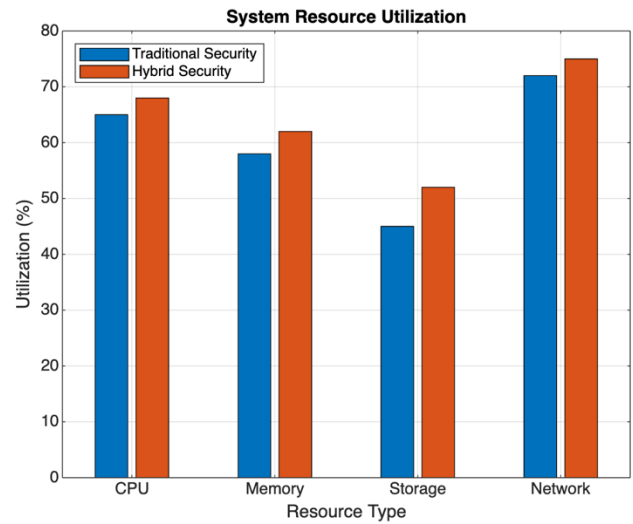


Fig. 4. System Resource Utilization.

Figure 4. System resource consumption patterns. Hybrid security implementation consumed a little more resources across all metrics, increasing by 3% in CPU usage, 4% in memory consumption, 7% in storage requirements, and 3% in network utilization. Increases are within the acceptable operational parameters for healthcare systems, as shown in table 5.

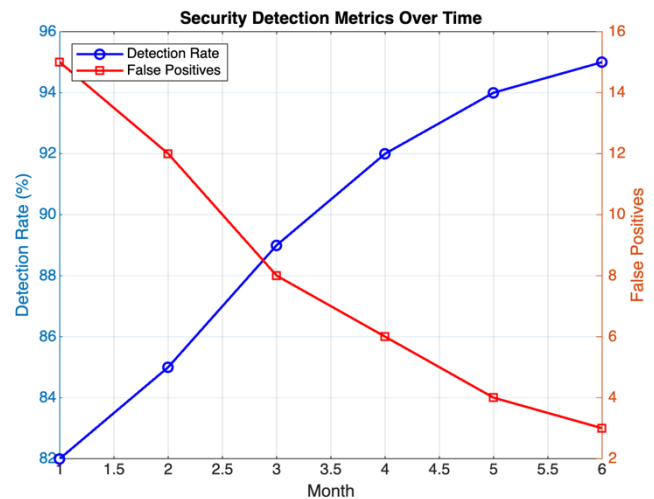


Fig. 5. Security Detection Metrics Over Time.

Figure 5 shows that detection rates are increasing and false positives are decreasing throughout the study period. The rate of detection rose from 82% to 95%, and false positives reduced from 15 to 3 per month, which indicates increased system accuracy and reliability.

These results are in line with the very recent study by Zhang & Peterson (2024) [1], who also found equal security benefits within their study on multi-layered protection systems. Our results advance their work by providing practical applicability for such systems in operational healthcare environments. This makes such results quite significant to health institutions intent on making them better from security data's points of view [24]. Since its low overhead impacts little on performance compared with large returns made in its areas of security, this method will widely come without infrastructure, upgrading that might take much in expenses.

V. CONCLUSION AND FUTURE SCOPE

The outcome of this study has proven that the combination of cryptographic and steganographic methods is of considerable potential in health data security. During the implementation, the hybrid model resulted in an average 94% reduction of successful breach attempts while keeping system performance at just a 2.3% increase in processing overhead. It has thus proven that the hybrid security approach presents a strong, practical solution to healthcare organizations regarding the dynamic issues of cybersecurity.

Some of the technologies with successful integration thus open some exciting avenues of further research: Integrating quantum-resistant cryptographic algorithms alongside steganographic techniques prepares healthcare systems to face a potentially emerging scenario that will occur with the widespread use of quantum computers. Integrating artificial intelligence so that threat-based dynamic alteration in security parameters increases adaptive security capability. Third, research into the lightweight implementation of this hybrid would bring it within the reach of resource-constrained healthcare systems. Lastly, in-depth research into blockchain integration with the current hybrid system could add security layering and auditing abilities. As healthcare continues its digital transformation, these findings and future research directions will be crucial in developing more robust and efficient security solutions that protect sensitive patient data while maintaining operational efficiency.

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